

1 · The problem

AI agents now spend money on people's behalf: about 1 in 6 Black Friday 2025 purchases were AI-assisted; generative-AI retail traffic grew 4,700% year over year (Adobe Analytics); Juniper Research (2026) names trust the #1 adoption barrier for agentic commerce. The rails agents pay on are instant and final by regulation — Pix (MED 2.0), SEPA Instant, FedNow, UPI — and the ECB puts instant-payment fraud risk at “up to 10× higher”. Dispute economics were already broken on cards: 106M disputes at Visa in 2025, 40–80% of e-commerce disputes classed as friendly fraud, roughly \$28B/yr in merchant chargeback losses. When an agent exceeds its mandate, pays the wrong party, or prepays for a service that never arrives — on a settlement-final rail, across borders where every rail has different dispute rules — there is no recourse layer. And as agents delegate to sub-agents and trade with agents they have never met, the authority chain has no enforcement.

2 · The market today — each product covers one part

Player / category	What it solves	What it doesn't
Visa AI dispute suite (2026)	Card dispute automation at network scale	Card-only, single network
Mastercard (Mastercom, Ethoca, Agent Pay)	Card disputes as APIs; intent standard	Protocols, not operations: no cross-rail orchestration, no escrow
Amex Agent Purchase Protection	Liability cover for agent purchases	Closed loop — Amex only
Agent-payment protocols (AP2, ACP, OKX APP, x402)	Agent checkout and payment execution	Dispute resolution absent or “future scope”
Chargeback SaaS	Card representment for merchants	Nothing for irrevocable A2A or agent mandates
Escrow / custody providers	Licensed fund-holding on specific rails	No programmable, dispute-aware, rail-agnostic escrow

3 · The Twist solution — four checks in fixed order

1 · Jurisdiction, agreed up front. Both parties rank the regions they accept; the service returns the Nash-bargaining optimum over 22 real regimes (Pix, SEPA, FedNow, UPI, FPS, NPP, M-Pesa, x402 stablecoin, ...) — the product of both parties' preference scores is maximised, so the pick rewards mutual acceptability. No shared region → null → no transaction. **2 · Consent, enforced.** The principal's mandate — budget and merchant allowlist — is stored before the agent pays; a payment outside it is refused with 403 before funds move. **3 · Escrow, delivery-gated.** Funds leave the payer at open and reach the payee only when the posted proof equals the agreed condition exactly; contest blocks release, refund makes the payer whole. **4 · Recall, rule-gated.** A settled payment reverses only where the regime allows recall, inside its window, on a proven mandate breach. **Grounded in law:** every dispute reason code maps to one of 7 civil/commercial-law categories; every jurisdiction lists its statutes, scheme rulebooks, and regulator with links to official sources; an accepted dispute returns the citation inline. **x402 on-rail:** 402 challenge → signed X-PAYMENT → resource + receipt; each nonce settles once; the consent cap is checked even on this irrevocable rail. **Machine-to-machine:** budget-bounded delegation chains (a sub-agent can never exceed its delegator) and one-call agent-to-agent pacts: Nash region + mandate check + funded escrow.

4 · The four trust relationships, one gate logic

Relationship	What is checked	Endpoints
Human → agent	payment stays inside the human's mandate	/consent → /pay
Agent → sub-agent	delegated budget ≤ delegator's budget	/m2m/delegate
Agent ↔ agent	shared region + mandate + escrow, one call	/m2m/handshake
Agent → paid resource	per-request payment; replay + budget checked	/x402/resource/{name}

5 · Proof

Adversarially validated: ships as a payments-layer plugin in MIT/NANDA's Nanda Town rig — 4 swarm scenarios, 10 adversarial validators; every attack (unilateral clawback, escrow drain, over-mandate spend, friendly-fraud refund, off-regime dispute) passes against the default ledger and is blocked by Twist on identical scenarios; 586 rig tests green. **Live service:** 33 endpoints, 61 endpoint tests, and a 41/41 live sweep on the deployed URL including stateful escrow, recall, dispute, m2m, and x402 flows — 647 tests green project-wide. **Dual-use site:** browsers get an interactive page (live playground, jurisdiction globe with per-region legal clauses, Nash 1–1/1–n/n–n cooperation visualizer, escrow builder, one-click 10-check API self-test); agents get JSON and SKILL.md at the same URL — every endpoint documented with one real call and its real response. **Flagship use case:** DigiDoot — a personal AI agent for every Indian citizen — consent → intent vault, UPI → regime, journeys → escrow/recall.

6 · Verify it yourself (60 seconds)

```
curl -s https://tvist-api-mias-projects-667d1349.vercel.app/health → {"status":"ok"}
curl -s -X POST https://tvist-api-mias-projects-667d1349.vercel.app/regions/recommend -d
'{"client_prefs":["eu_sepa","br_pix"],"agent_prefs":["br_pix","in_upi"]}' → agreed_region "br_pix"
open https://tvist-api-mias-projects-667d1349.vercel.app → click "Test the API" — 10 live checks run in the page.
```

Live: <https://tvist-api-mias-projects-667d1349.vercel.app> · agent contract:

<https://tvist-api-mias-projects-667d1349.vercel.app/skill.md> · source: github.com/superhero888/tvist-api

Technical demonstration only — a notional-credits sandbox, not a bank, payment institution, or law firm. Legal references cite real primary sources but are engineering-grade mappings, not legal advice; obtain qualified local counsel before relying on them in any jurisdiction. Market figures from the named public sources.